

Traditional Web



Traditional Web



Single Page Application(SPA)



A Single Page Application (SPA) is a web application that loads a single HTML page and dynamically updates its content as the user interacts with the app. This means that instead of loading entirely new pages from the server, SPAs use JavaScript to update the page content without requiring a full page refresh.

Visual Studio Code interface showing a React project named SPA-6. The Explorer sidebar on the left displays the file structure:

- SPA-6
 - pages
 - Course.jsx
 - Home.jsx
 - App.css
 - App.jsx
 - index.css
 - main.jsx
 - .eslintrc.cjs
 - .gitignore
 - index.html
 - package-lock.json
 - package.json
 - README.md
 - vite.config.js
- OUTLINE
- TIMELINE

The Home.jsx file is open in the editor, showing the following code:

```
1 import {Link} from "react-router-dom";
2 const Home =()=>{
3   return (
4     <>
5     <h1>Home Page</h1>
6     <Link to="/course">Course</Link>
7     </>
8   )
9 }
10
11 export default Home;
```

The bottom panel shows the TERMINAL output:

```
9:27:05 PM [vite] hmr update /src/pages/Home.jsx (x2)
```

A small window titled "vulnerabilities" is visible in the bottom right corner of the terminal area, showing a list of vulnerabilities for the project.

Host

Course Page

SPA-6

EXPLORER

SPA-6

- pages
 - Course.jsx
 - Home.jsx
- # App.css
- App.jsx 2
- # index.css
- main.jsx
- .eslintrc.cjs
- .gitignore
- index.html
- package-lock.json
- package.json
- README.md
- JS vite.config.js

App.jsx 2

```
1 import {RouterProvider,createBrowserRouter} from "react-router-dom"
2 import Home from "../pages/Home"
3 import Course from "../pages/Course"
4 import './App.css'
5
6 const router = createBrowserRouter([
7   {path:"/",element:<Home/>},
8   {path:"/courses", element:<Course/>}
9 ])
10 function App() {
11
12   return (
13     <>
14     <Home></Home>
15   )
16 }
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

9:22:32 PM [vite] hmr update /src/App.jsx

found 0 vulnerabilities

PS C:\Users\user\Desktop\React\SPA-6\client>

Ln 9, Col 3 Spaces: 2 UTF-8 LF { } JavaScript

SPA-6

EXPLORER

SPA-6

- pages
 - Course.jsx
 - Home.jsx
 - # App.css
 - App.jsx 2
 - # index.css
 - main.jsx
- .eslintrc.cjs
- .gitignore
- index.html
- package-lock.json
- package.json
- README.md
- JS vite.config.js

client > src > App.jsx > ...

```
1 import {RouterProvider,createBrowserRouter} from "react-router-dom"
2 import Home from './pages/Home.jsx'
3 import './App.css'
4 function App() {
5   return (
6     <Home></Home>
7   )
8 }
9
10 export default App
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

9:19:47 PM [vite] hmr update /src/pages/Home.jsx (x2)

found 0 vulnerabilities

PS C:\Users\user\Desktop\React\SPA-6\client>

Ln 1, Col 56 Spaces: 2 UTF-8 LF JavaScript

SPA-6

EXPLORER

- SPA-6
 - client
 - node_modules
 - public
 - src
 - assets
 - pages
 - Course.jsx
 - Home.jsx
 - App.jsx**
 - App.css
 - index.css
 - main.jsx
 - .eslintrc.cjs
 - .gitignore
 - index.html
- OUTLINE
- TIMELINE

App.jsx

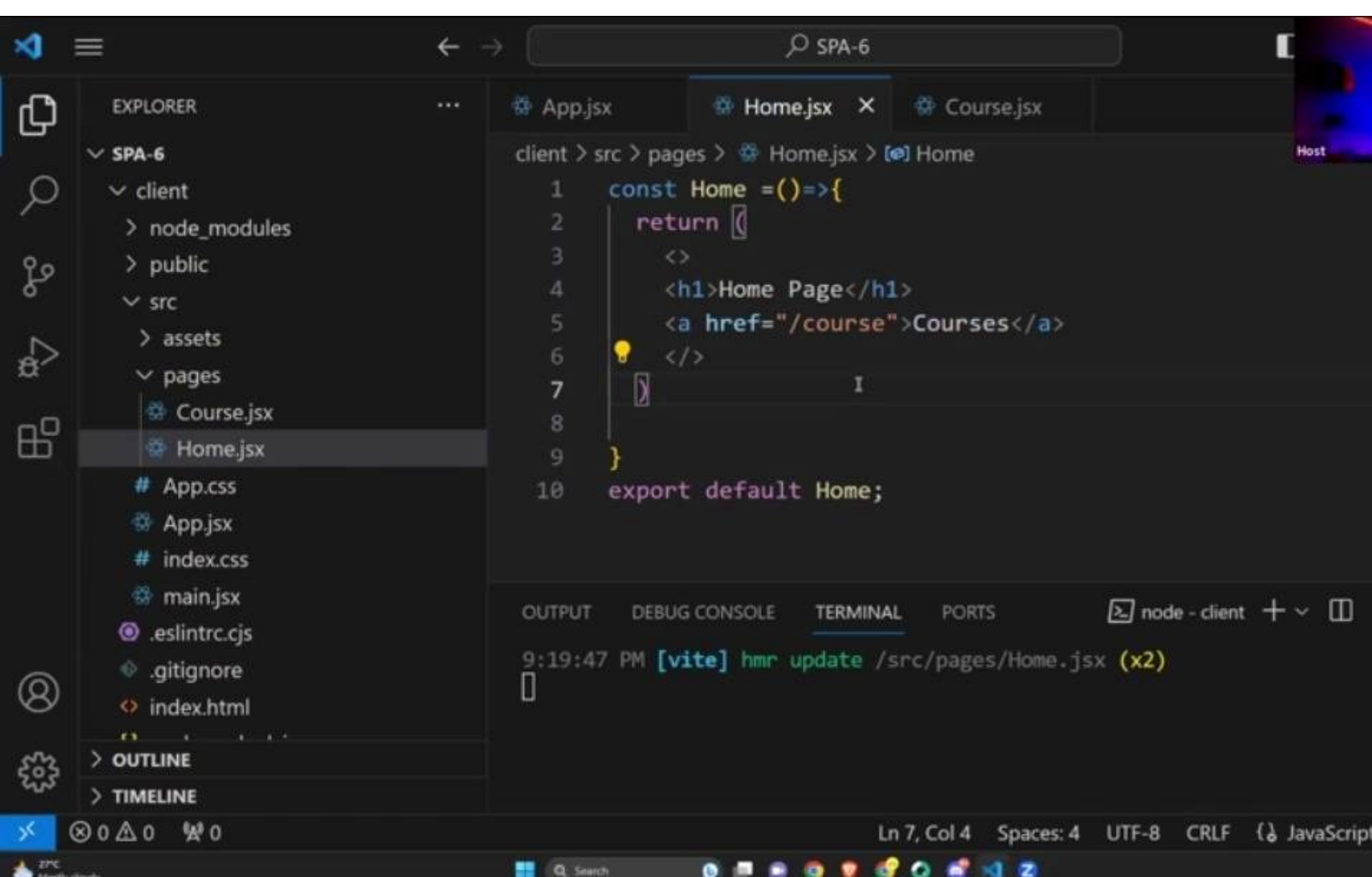
```
client > src > App.jsx > ...
1  import Home from "../pages/Home"
2  import './App.css'
3  function App() {
4
5      return (
6          <>
7              <Home></Home>
8          </>
9      )
10 }
```

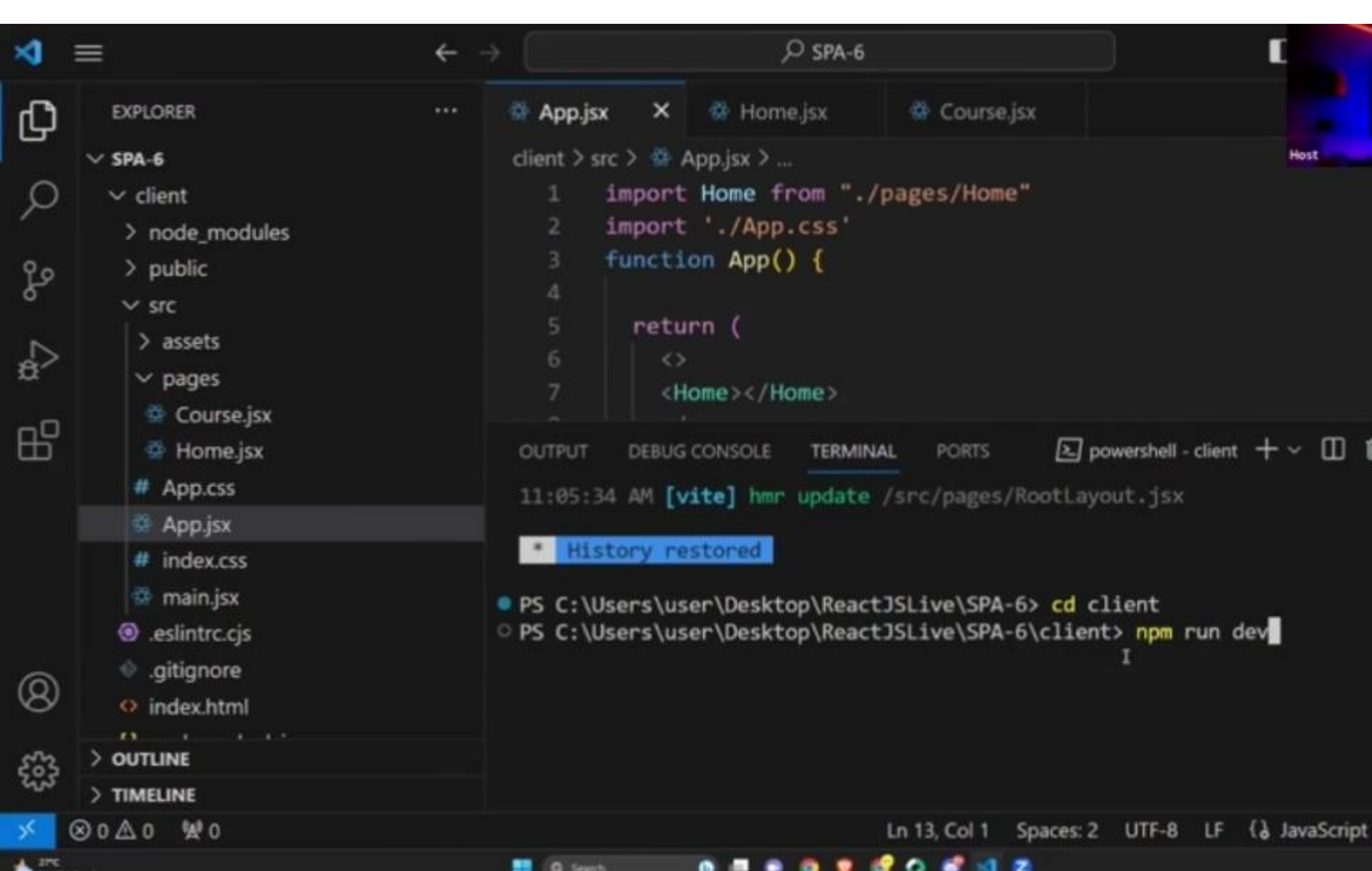
OUTPUT DEBUG CONSOLE TERMINAL PORTS

9:19:47 PM [vite] hmr update /src/pages/Home.jsx (x2)

PS C:\Users\user\Desktop\React\SPA-6\client> npm install react-router-dom

Ln 13, Col 1 Spaces: 2 UTF-8 LF JavaScript





SPA-6

EXPLORER

SPA-6

client

node_modules

public

src

assets

pages

Course.jsx

Home.jsx

App.css

App.jsx

index.css

main.jsx

.eslintrc.cjs

.gitignore

index.html

OUTLINE

TIMELINE

App.jsx

Home.jsx

Course.jsx

client > src > App.jsx > App

1 import Home from "../pages/Home"

2 import './App.css'

3 function App() {

4

5 return (

6 <>

7 <Home> / Home

8 </>

9)

10 }

11

12 export default App

13

Ln 7, Col 11

Spaces: 2

UTF-8

LF

{ } JavaScript

SPA-6

EXPLORER

- SPA-6
 - client
 - node_modules
 - public
 - src
 - assets
 - pages
 - Course.jsx
 - Home.jsx
- App.css
- App.jsx
- index.css
- main.jsx
- .eslintrc.cjs
- .gitignore
- index.html

OUTLINE

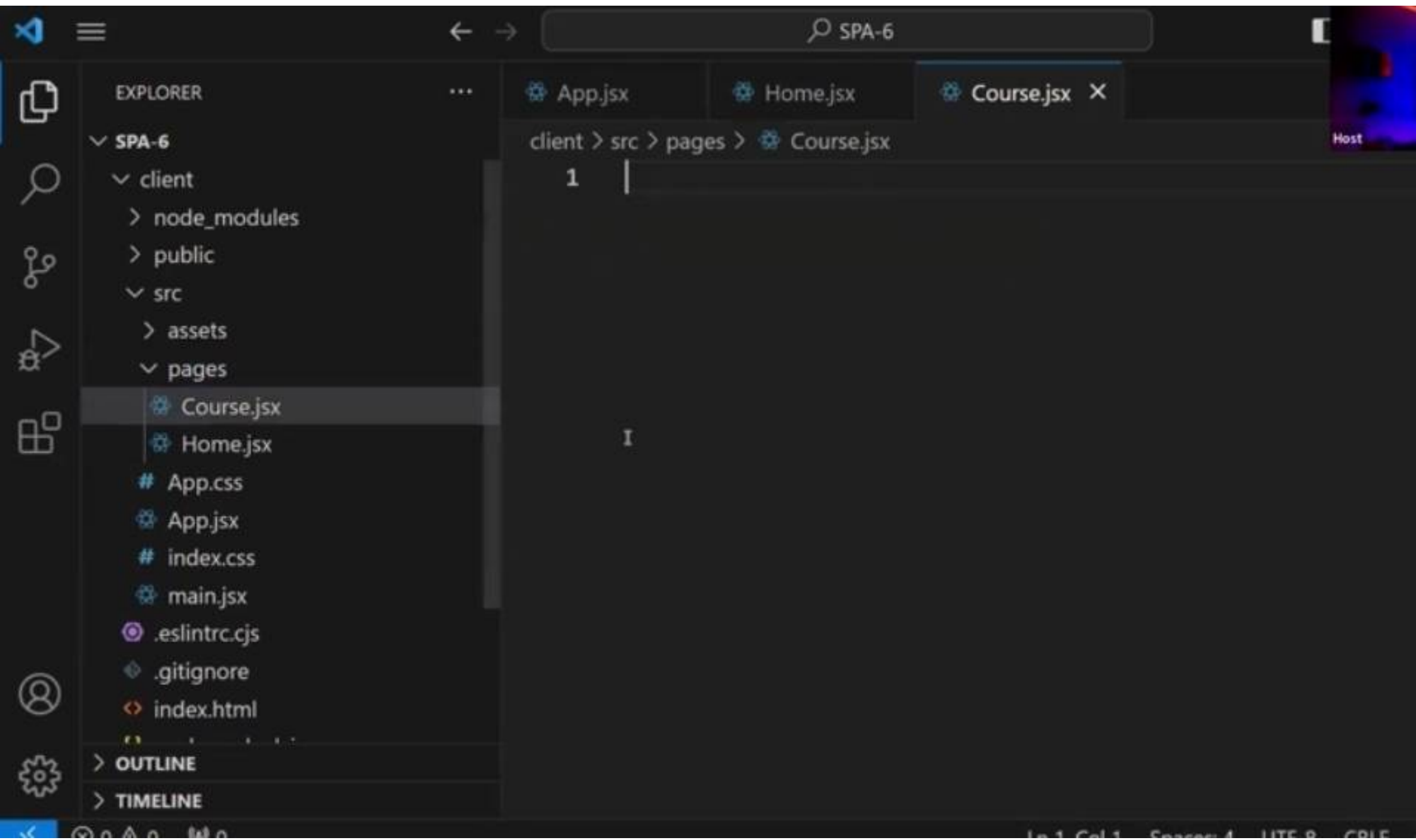
TIMELINE

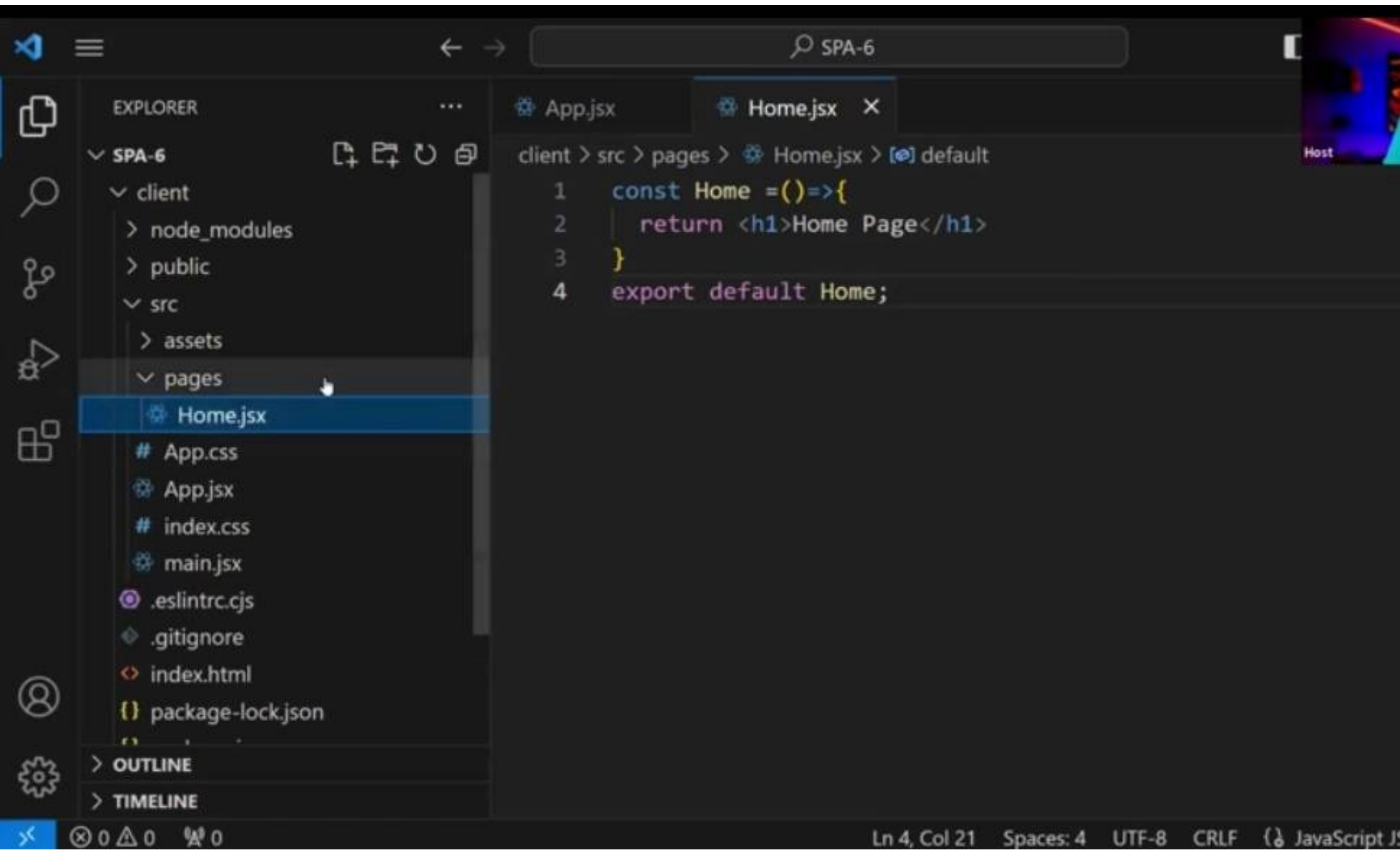
App.jsx Home.jsx Course.jsx

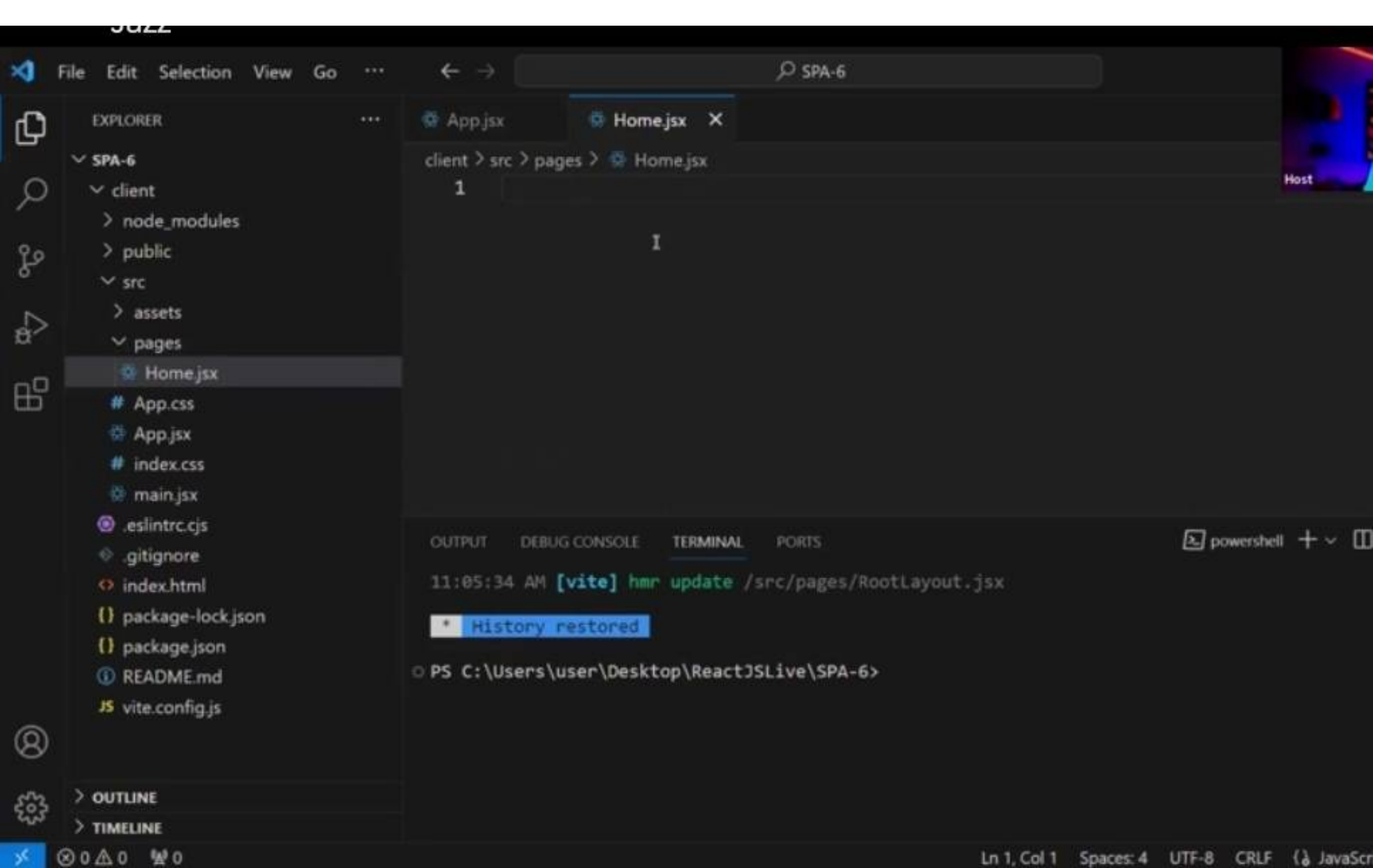
client > src > pages > Course.jsx > Course

```
1 const Course = () => {  
2   return <h1>Course Page</h1>  
3 }  
4 export default Course;
```

Ln 2, Col 22 Spaces: 4 UTF-8 CRLF JavaScript







File Edit Selection View Go ... SPA-6

EXPLORER

- SPA-6
 - client
 - node_modules
 - public
 - src
 - pages
 - assets
 - App.css
 - App.jsx
 - index.css
 - main.jsx
 - .eslintrc.cjs
 - .gitignore
 - index.html
 - package-lock.json
 - package.json
 - README.md
 - vite.config.js

App.jsx

```
client > src > App.jsx > App
1 import './App.css'
2 function App() {
3
4   return (
5     <>
6     <h1>Hi</h1>
7     </>
8   )
9 }
10
11 export default App
12
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

11:05:34 AM [vite] hmr update /src/pages/RootLayout.jsx

History restored

PS C:\Users\user\Desktop\ReactJSLive\SPA-6>

Ln 4, Col 11 Spaces: 2 UTF-8 LF JavaScript

File Edit Selection View Go ... SPA-6

EXPLORER

- SPA-6
 - client
 - node_modules
 - public
 - src
 - .eslintrc.cjs
 - .gitignore
 - index.html
 - package-lock.json
 - package.json
 - README.md
 - vite.config.js

App.jsx

```
client > src > App.jsx > App
1 import './App.css'
2 function App() {
3
4   return (
5     <>
6     <h1>Hi</h1>
7     </>
8   )
9 }
10
11 export default App
12
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

11:05:34 AM [vite] hmr update /src/pages/RootLayout.jsx

* History restored

PS C:\Users\user\Desktop\ReactJSLive\SPA-6>

Ln 4, Col 11 Spaces: 2 UTF-8 LF JavaS